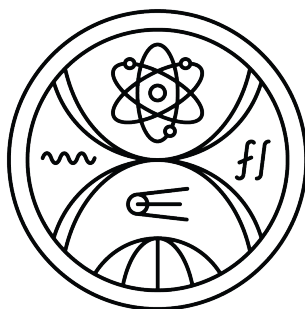


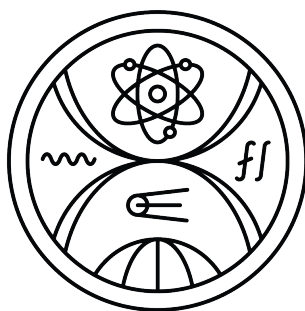
COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



AUTOMATED EVALUATION OF THE REY-OSTERRIETH COMPLEX IMAGE TEST

Master's thesis

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Master's thesis

Study program: Applied Informatics
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Department: Department of Applied Informatics
Supervisor: doc. RNDr. Zuzana Černeková, PhD.



Univerzita Komenského v Bratislave
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ZADANIE ZÁVEREČNEJ PRÁCE

- Meno a priezvisko študenta:** Bc. Lucia Korbeľová
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- Názov:** Automatizácie hodnotenie Rey-Osterriethovho komplexného obrázkového testu
Automated evaluation of the Rey-Osterrieth complete image test
- Anotácia:** Poruchy pamäti sú charakteristickým znakom mnohých rôznych neurologických a psychiatrických ochorení. Rey-Osterriethova komplexná figúra (ROCF) je najmodernejším hodnotiacim nástrojom neuropsychológov na celom svete na posúdenie stupňa zhoršenia neverbálnej vizuálnej pamäte. Na získanie skóre vyškolený klinický lekár kontroluje kresbu ROCF pacienta a kvantifikuje odchýlky od pôvodnej kresby.
- Cieľ:** Cieľom práce je vytvoriť automatizovaný systém pre hodnotenie neuropsychologického testu Rey-Osterriethovej komplexnej figúry (ROCF) s použitím metód počítačového videnia. Tento systém bude využívať pokročilé techniky počítačového videnia a umelej inteligencie na automatické rozpoznávanie a kvantifikáciu odchýlok od pôvodnej kresby. Účinnosť systému a jeho presnosť budú porovnávané s tradičnými metódami hodnotenia. Cieľom je poskytnúť efektívny nástroj na automatizované hodnotenie neuropsychologických testov, čo môže prispieť k lepšiemu diagnostikovaniu a monitorovaniu neurologických a psychiatrických porúch pamäte.
- Literatúra:** Davide Di Febbo, Simona Ferrante, Marco Baratta, Matteo Luperto, Carlo Abbate, Pietro Davide Trimarchi, Fabrizio Giunco, Matteo Matteucci; A decision support system for Rey–Osterrieth complex figure evaluation; Expert Systems with Applications, Volume 213, Part C, 2023, 119226, ISSN 0957-4174, <https://doi.org/10.1016/j.eswa.2022.119226>. (<https://www.sciencedirect.com/science/article/pii/S0957417422022448>)
- Park, J.Y., Seo, E.H., Yoon, HJ. et al. Automating Rey Complex Figure Test scoring using a deep learning-based approach: a potential large-scale screening tool for cognitive decline. Alz Res Therapy 15, 145 (2023). <https://doi.org/10.1186/s13195-023-01283-w>
- R. O. Canham, S. L. Smith and A. M. Tyrrell, "Automated scoring of a neuropsychological test: the Rey Osterrieth complex figure," Proceedings of the 26th Euromicro Conference. EUROMICRO 2000. Informatics: Inventing the Future, Maastricht, Netherlands, 2000, pp. 406-413 vol.2, doi: 10.1109/EURMIC.2000.874519.



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Fakulta matematiky, fyziky a informatiky

Vedúci: doc. RNDr. Zuzana Černeková, PhD.
Katedra: FMFI.KAI - Katedra aplikovanej informatiky
Vedúci katedry: doc. RNDr. Tatiana Jajcayová, PhD.
Dátum zadania: 19.10.2023

Dátum schválenia: 14.11.2023

prof. RNDr. Roman Ďurikovič, PhD.
garant študijného programu

.....
študent

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vedúci práce

I hereby declare that I have independently completed this master's thesis on the topic 'Automated Evaluation of the Rey-Osterrieth Complex Image Test', including all its appendices and images, using the literature listed in the attached bibliography and artificial intelligence tools, under the careful supervision of my thesis advisor. I declare that I have used artificial intelligence tools in accordance with applicable legal regulations, academic rights and freedoms, ethical and moral principles, while maintaining academic integrity, and that their use is appropriately indicated in the work.

Bratislava, 2026

.....
Bc. Lucia Korbeľová

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Abstrakt

S narastajúcim počtom ľudí trpiacich následkami demencie a iných neuropsychologických porúch narastá aj potreba spoľahlivých a efektívnych diagnostických nástrojov. Jednou z najčastejšie používaných metód na hodnotenie vizuálnej pamäti a kognitívnych schopností je test Rey–Osterriethovej komplexnej figúry (ROCF). Manuálne hodnotenie tohto testu je však časovo náročné a subjektívne, čo do určitej miery obmedzuje jeho častejšie využitie v klinickej praxi. Z tohto dôvodu sa naša práca sústreďuje na vytvorenie automatizovaného systému, ktorý je schopný hodnotiť ROCF kresby jednotným a objektívnym spôsobom s využitím moderných metód počítačového videnia a hlbokého učenia. V tejto diplomovej práci porovnávame dve architektúry hlbokého učenia, konvolučnú neurónovú sieť ResNet18 a Swin Transformer, v zjednodušenom klasifikačnom a trénovacom nastavení, aby sme určili, ktorá z nich dosahuje lepšie výsledky a ktorá z piatich augmentačných techník prináša najvyššiu presnosť. Analýza ukázala, že ResNet18 dosiahol vyššiu presnosť a potvrdil, že je vhodnejší pre tento typ dát. Pomocou najlepšej trénovacej konfigurácie z tohto experimentu sme následne rozšírili náš prístup na šesťtriednu klasifikáciu, kde každá trieda predstavuje interval šiestich bodov v ROCF skórovacej škále, a ďalej sme optimalizovali techniky predspracovania pre tento model. Nakoniec sme predostreli a otestovali náš vlastný návrh kombinujúci kóder, ktorý extrahuje nízkoúrovňové príznaky, s jednoduchou neurónovou sieťou vykonávajúcou finálnu klasifikáciu, čo nám umožnilo preskúmať potenciál hybridnej architektúry založenej na kóderoch pre automatické skórovanie ROCF.

Kľúčové slová: Rey-Osterriethova komplexná figúra, hlboké učenie, konvolúčna neuronová sieť, transformer, kóder

Abstract

With the growing number of people suffering from dementia and other neuropsychological disorders, there is an increasing need for reliable and efficient diagnostic tools. One of the most widely used methods for assessing visual memory and cognitive abilities is the Rey–Osterrieth Complex Figure Test (ROCF). However, its manual evaluation is time-consuming and subjective, which limits its frequent use in clinical practice. For this reason, our work focuses on developing an automated system capable of evaluating ROCF drawings in a unified and objective way using modern computer vision and deep learning methods. In this thesis, we compare two deep learning architectures, convolutional neural network ResNet18 and Swin Transformer, in a simplified classification and training setting to determine which performs better and which of five augmentation techniques brings the best results. The analysis showed that ResNet18 achieved higher accuracy and proved to be more suitable for this type of data. Using the best training configuration from this experiment, we extended our approach to a six-class classification, where each class represents an interval of six points in the ROCF scoring scale, and further optimized preprocessing techniques for this model. Finally, we proposed and tested our own design combining an encoder that extracts low-level image features with a simple neural network that performs the final classification, allowing us to explore the potential of hybrid encoder-based architectures for ROCF scoring.

Keywords: Rey-Osterrieth complex figure, deep learning, convolutional neural networks, transformer, encoder

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Terminology

Terms

- **Some term**
Explanation of the term.

Abbreviations

- **ROCF** - Rey-Osterrieth complex figure.
- **CNN** - Convolutional neural network.
- **BQSS** - the Boston Qualitative Scoring System.
- **MCI** - Mild cognitive impairment
- **MAE** - Mean absolute error
- **ReLU** - Rectified Linear Unit
- **CV** - Cross validation.

Motivation

A world-renowned Alzheimer’s disease organization estimated in its World Alzheimer Report 2015 that “46.8 million people worldwide were living with dementia in 2015. This number will almost double every 20 years, reaching 74.7 million in 2030 and 131.5 million in 2050” [14]. The main reason, for this outlook is the increasing aging of the population. We expect that in the future there will increasing need for reliable diagnostic and progress assessment tools. One such diagnostic method is the pen-and-paper test known as the Rey–Osterrieth Complex Figure Test.

One such diagnostic method is the pen-and-paper test known as the Rey–Osterrieth Complex Figure Test. This neuropsychological test also has countless other applications in clinical practice and in research. It can be used as an effective tool for assessing visual memory, visuo-constructional abilities, fine-motor skills, and organizational and planning capabilities. Additionally, it was originally designed to monitor recovery progress during treatment of a brain injury and is often used to help diagnose various neurological and psychiatric disorders, including, but not limited to dementia, Parkinson’s disease, schizophrenia, and depression.

However, there are multiple issues connected to the ROCF test that have led to a gradual decline in its usage despite its wide range of applications [3]. One of the problems is that administering the test and performing its subsequent evaluation takes a substantial amount of time. In addition, the scoring process is considered to be subjective. It involves assessing the similarity between drawn and reference shapes, therefore the final scores can vary from one expert to another. Meaning the possibility of standardization of the scores is limited [12]. For these reasons, the need to develop an automated scoring system for the ROCF test has emerged, as a way to ensure a unified scoring procedure and significantly faster evaluation.

There have been numerous attempts by researchers to create such an automated scoring system for the ROCF test. Early attempts focused on using classical computer vision algorithms and pattern recognition methods. They mainly tried to identify simple sub-elements of the drawings and then evaluate their similarity to the reference template. However, from our literary review, we believe that approaches using solely classical computer vision are not very likely to be successful. We noticed that the biggest obstacle for these algorithmic approaches was the high level of variation, dis-

tortion, and rearrangement of elements in the drawings made by patients [4]. Because of this, early systems had very limited functionality and were only able to successfully recognize and score a small number of the simplest patterns in the ROCF. These patterns included only rectangles and triangles. One such system was created by Canham, Smith, and Tyrrell in 2000 [4].

In our opinion, the use of machine learning and deep learning methods has a great potential to improve the quality of software in this field. These techniques can effectively process complex images and automatically discover important discriminative features. Although there have been several promising attempts to build automated ROCF scoring systems abroad, there is still no practical or usable software designed specifically for Slovakia that follows the Slovak standardization of ROCF test evaluation. Our thesis aims to contribute towards developing such a system.

Our goal is to create an automated scoring model for ROCF drawings that is able to approximate the drawing score in six-point intervals, categorizing the images into six groups with a similar level of quality. To achieve this, we compare several preprocessing and augmentation techniques. We also evaluate different types of machine learning models, including standard convolutional neural networks, transformer architectures, and encoder–neural network combinations.

Chapter 1

Introduction

The aim of the following chapter is to present the theoretical background of the Rey–Osterrieth complex figure test and to describe various existing methods that focus on automating the evaluation or diagnosis formulation based on Rey–Osterrieth complex figures. These previous studies have inspired the research methods used in this work. The first subchapter explains the history and various clinical applications of the ROCF Test, as well as alternative versions of complex figures that serve as equivalent or simplified alternatives to the original ROCF. Finally, it briefly introduces different types of scoring systems used to comprehensively evaluate cognitive functions of a subject, with a focus on Osterrieth’s scoring system, which was used to produce the scores in our dataset.

In the subchapter on existing methods, we summarize two research papers by research groups from abroad, one combining several advanced computer vision and deep learning techniques to predict patient diagnoses, and the other using a purely deep learning approach to estimate ROCF scores from drawings. Additionally, we present two recently completed master’s theses from Slovakia that worked with the same image dataset used in this thesis.

1.1 Rey-Osterrieth complex figure test

Clinical neuropsychology is defined as a scientific discipline within psychology that focuses on studying the impact of neurological dysfunctions and injuries on an individual’s psychological processes and behavior [2]. One of the main goals of neuropsychology is to diagnose brain dysfunctions by analyzing nonstandard and suboptimal behavior and cognitive performance [12]. For this purpose, various assessment methods are used, including ‘pen-and-paper’ tests, during which the subject writes or draws on paper. These types of tests are widely used because drawing and writing are complex graphomotor tasks that require visual-perceptual, motor, and coordination skills, which are

highly sensitive to brain disorders affecting these abilities [12]. One commonly used test of this type is the Rey-Osterrieth complex figure test (ROCF). It is classified as a reproduction task, which means that the participant is asked to copy or recall from memory a geometric figure shown to them.

The Rey-Osterrieth complex figure is a standardized illustration that consists of multiple complicated geometric shapes, as shown in Figure 1.1 [16]. The ROCF test, on which we are focusing in our thesis, can be divided into three stages. Each stage producing an independent and unique version of the figure drawn by the participant. The first stage, called the Copy phase, requires the individual to reproduce the ROCF image, which is presented in front of them, as accurately as possible while viewing the template of the desired figure. Once this drawing is completed, the template is hidden from the subjects view, and the participant is asked to draw the same figure, but this time entirely from memory, this is known as the Immediate Recall phase. The final stage, Delayed Recall, takes place after approximately additional 30 minutes. The subject is once again asked to recall and redraw the figure. The evaluator then scores each of the three drawings based on how much they resemble the original figure. Additionally, depending on the scoring system used, the evaluator can also take into account the participant's planning and drawing strategy [15].

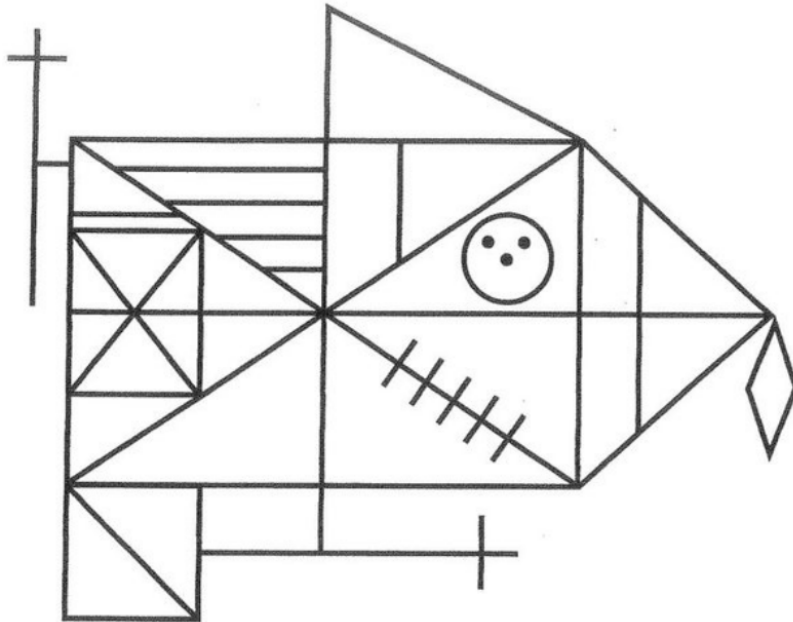


Figure 1.1: Rey-Osterrieth complex figure, Source: [5].

The origins of the Rey-Osterrieth complex figure test date back to 1941, when André Rey introduced the Complex figure test as an effective method for assessing visual memory and visuospatial constructional ability of patients who suffered a brain

injury, to assess the extend of their injuries. This influential work was then followed up by Paul-Alexandre Osterrieth in 1944 who standardized the way the test is performed and its scoring system, which is commonly used to this day [15]. Since its introduction, the test has been extensively applied in both research and clinical contexts for numerous different purposes.

Cognitive functions typically assessed and measured with the ROCF test in both both research setting and medical practice include nonverbal and visual memory, that is, the ability to remember visual information, as well as the ability to concentrate on the given task, fine-motor skills, the ability to understand and process the spatial relationship of objects in an image, organization and planning abilities, among others [16]. Naturally, a subject's performance in the Copy and Recall phases of the test can reveal different types of neuropsychological dysfunctions. The two Recall tasks mainly evaluate visuospatial memory, whereas the Copy task provides insight into the ability to maintain attention, visuospatial perception required to break down the figure into its components and identify their shapes and positions, and the ability to organize and plan the drawing based on these subunits [15].

Psychologists, that are professionally trained are able to use the results of the evaluation to non-invasively recognize and diagnose numerous neuropsychological disorders and brain dysfunctions, including dementia, Alzheimer's disease, Parkinson's disease, attention deficit hyperactivity disorder (ADHD), eating disorders, depression, schizophrenia and the effects of traumatic brain injury [16].

1.1.1 Alternative Versions of the Rey–Osterrieth complex figure

Over time, since the original development of the Rey–Osterrieth complex figure, several alternative versions of complex figures have been designed for similar testing purposes. These variants were created either to provide figures of comparable difficulty to the original ROCF as an alternative, or to simplify the task for special circumstances, for example, for different age or education groups. Examples of these versions are shown in Figure 1.2.

Initially, the reason for developing numerous figures was related to the original purpose of the ROCF test (Figure 1.2.A), which was used before and after treatment of a brain injury as an assessment tool to evaluate deterioration or improvement of visuospatial capabilities after treatment. However, that meant that one patient might be asked to perform the ROCF test with the same figure multiple times. This may cause the subject to perform better simply because they remember the figure from previous trials. This is negatively affecting the reliability of the results.

To address this issue, researchers designed several new complex figures that are

equally difficult to reconstruct and recall as the original Rey–Osterrieth complex figure [16]. The first one was the Taylor figure (Figure 1.2.B), introduced in 1969, which was thought to be equivalent. However, after subsequent research it was found to be interchangeable with the ROCF in evaluation of copying ability but not when it comes to assessing nonverbal memory and organizational skills [9]. To improve equivalence, the modified Taylor figure (Figure 1.2.C) was developed, which is slightly modified version of Taylor figure with higher complexity which was proved to be truly equivalent in all aspects with ROCF. Other figures with comparable properties are the Mark figure (Figure 1.2.D) and the Medical College of Georgia Complex Figures (Figures 1.2.E–H) [16].

Furthermore, it has been shown that the additional disadvantage of ROCF is that it's score increases with the higher levels of education achieved by the individuals and declines with increasing age [7], which makes the test a less reliable for elderly or low-educated populations. Therefore, various simplified versions of the ROCF and Taylor figure were created to ensure valid assessment of these groups (Figures 1.2.I–L) [16].

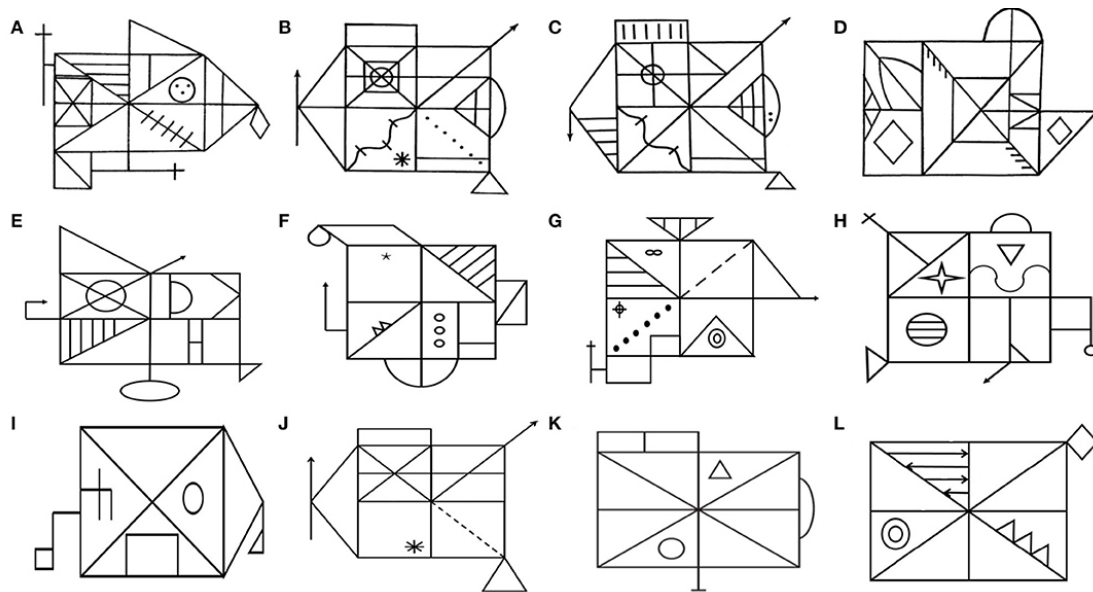


Figure 1.2: Example of the alternative versions of Rey–Osterrieth complex figure, the original Rey–Osterrieth complex figure (A), the Taylor figure (B), the modified Taylor figure (C), the Mark figure (D), the Medical College of Georgia Complex Figure (E–H), the Benson figure (I), the simplified Taylor figure (J), the simplified versions of ROCF (I–L), Source: [16].

1.1.2 Scoring Systems and Evaluation Methods

Over the years, multiple scoring systems and methodologies have been developed to assess the Rey–Osterrieth complex figure and other similar complex figures. Each scoring system was designed for a specific research or clinical purpose, focusing on

different cognitive processes. Although each approach has its particularities, two main categories of scoring systems can be distinguished. The first category includes accuracy based scoring methods, such as Osterrieth’s original scoring system, the Meyers scoring system, and others. The second category consists of process based scoring systems, including the Boston Qualitative Scoring System (BQSS) and the Q-Score [16].

Accuracy-based scoring systems are simpler, more straightforward and generally more reliable and reproducible. Which makes them more practical for clinical use. Their main focus is to evaluate visuo-constructional abilities and nonverbal memory. The result of these methods is a quantitative score that represents how closely the subject’s drawing resembles the original template figure. The figure is divided into smaller logical units, and each element is scored based on how similar the shape is to the original and whether it is placed correctly. However, a limitation of these quantitative methods is that they only consider the final static drawing and ignore the drawing process itself. As a result, valuable information is completely lost, such as the order in which elements were drawn and how the subject organized the figure into subunits [16].

Process-based scoring methods were developed to address the limitations of the accuracy-based approach. They aim to more accurately assess cognitive and executive functions that are not captured by accuracy scoring methods, including organizational strategy, planning (drawing order), direction, and placement of elements in the figure. Although process scoring systems provide a more detailed and thorough evaluation of cognitive abilities, they are less frequently used in clinical settings because they require a significant amount of time to complete and the evaluation process is excessively complex and time-consuming [16].

The BQSS is an example of a process-based scoring methodology. It produces multiple scores that each reflect different aspects of cognitive abilities [15]. The system evaluates not only accuracy and correctness of placement but also planning, size distortion, rotation, asymmetry, and other qualitative characteristics [16]. To capture the drawing order and assess organizational and planning abilities, multiple colored markers are often used to distinguish between different phases of the drawing process [15].

1.1.3 Osterrieth’s Scoring System

Osterrieth’s accuracy scoring system was developed in 1944 by Paul Osterrieth as the first scoring methodology for the Rey–Osterrieth complex figure test, and it remains to be commonly used today [9]. The main focus of this evaluation method is to assess the correctness of placement and the geometrical accuracy of specific sub-elements of the ROCF drawing. Therefore this evaluation is able to effectively quantify visual memory

and visuo-spatial abilities, however, like other accuracy-based scoring methods, it is incapable of assessing organizational or executive functions [15].

In his original work, Osterrieth divided the Rey–Osterrieth complex figure into 18 independent subunits (see Figure 1.3). Each of these 18 elements is either a simple geometric shape, such as a rectangle, triangle, rhombus or line, or a logical grouping of geometric figures, such as a circle with three dots, a cross, or multiple parallel lines [16].

The scoring procedure is relatively simple and can be divided into two main phases. The first step of the evaluation is to divide the subject’s drawing into the 18 elements as defined in Figure 2. However, this process is not as straightforward as it might seem, since not all of the original 18 units are typically present in the subject’s drawing. Some elements might be missing and those that are present are often deformed, incomplete, or placed in unexpected positions. As a result, this division step can differ from expert to expert [16].

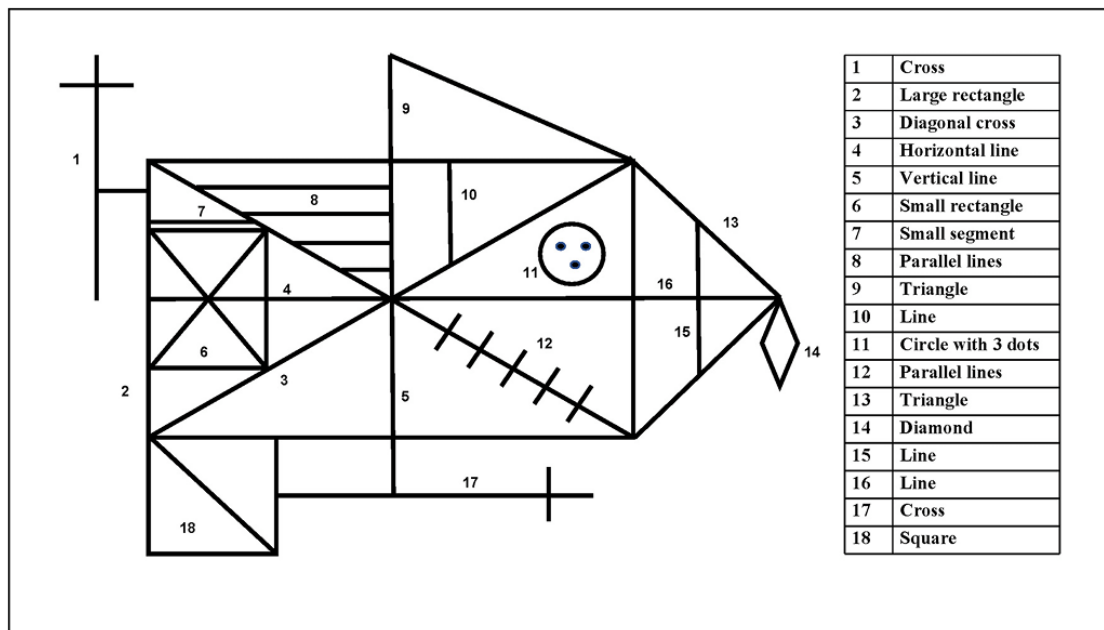


Figure 1.3: Definition of 18 scoring units of Rey–Osterrieth complex figure, Source: [16].

Subsequently, each of the 18 elements is scored independently and can receive up to two points, depending on whether the element is present in the drawing, accurately shaped, and correctly placed. There are five possible situations of how one element was drawn, and each of them has a specific score defined in Table 1.1. The element receives the full 2 points if it is present, accurately drawn, and correctly placed. Elements that are accurately shaped but misplaced, or correctly placed but have deformed shape, receive 1 point. If the element is both misshapen, but recognizable and misplaced, it receives 0.5 points. Naturally, omitted elements receive 0 points. Consequently, each

individual ROCF drawing can receive a total score ranging from 0 to 36 points [16].

When a subject receives a low score in the copy phase it indicates impaired brain function related to visuo-constructional abilities. If a low score was received in the immediate or delayed recall phases it suggests that the individual has suboptimal visual memory. More straightforwardly, it means that the drawn image is not very similar to the original ROCF template. On the other hand, healthy individuals are expected to achieve higher scores, as their drawing should more closely match the ROCF image, although the benchmark values also depend on age and educational background.

Score	Accuracy	Placement
2	Accurate shape	Correct placement
1	Accurate shape	Incorrect placement
1	Inaccurate shape	Correct placement
0.5	Inaccurate shape, but recognizable	Incorrect placement
0	Inaccurate shape and unrecognizable, or omitted	Incorrect placement

Table 1.1: Scoring criteria for the elements of ROCF drawing [15].

1.2 Existing methods

1.2.1 A decision support system for rey–osterrieth complex figure evaluation

The main goal of this study, published in 2023, was to develop a decision support system which could help clinicians with evaluating drawings produced during the copy phase of the Rey–Osterrieth complex figure test. Their dataset consisted of 250 copies of the ROCF from individuals who were either cognitively healthy, had mild cognitive impairment (MCI), or had dementia.

The proposed system generated two types of outputs. First, it assigned one of four qualitative labels to each of the 18 ROCF elements, representing specific error types instead of using the standard quantitative score between 0 and 2 typical for the Osterrieth’s scoring method. This reformulation simplified the scoring task into a classification problem of four classes. The labels were: omitted (element not present), distorted (shape recognizable but inaccurate), misplaced (correct shape but wrong location), and correct. The second output represented the individual’s overall diagnosis (healthy, MCI, or dementia) based on the qualitative labels predicted for each of the 18 elements.

The system’s workflow had three main phases. The first phase was image pre-processing, which included noise reduction using low-pass and median filtering, binarization through unimodal thresholding, and stroke enhancement using erosion, skele-

tonization, and dilation. The final preprocessing step involved figure standardization via image homography, which aligned the drawing's size, rotation, and proportions with the reference ROCF template using five manually selected key points of the main structure. These points were located at the four corners of the main rectangle (pattern 2, see Figure 1.4) and at the rightmost vertex of the main triangle (pattern 18, see Figure 1.4) [6].

The second phase, called pattern evaluation, focused on classifying each of the 18 elements of the ROCF into one of four error categories. These elements were divided into two types of patterns, simple and complex (see Figure 1.4). Simple patterns included lines, triangles, rectangles, and other polygons. Standard computer vision techniques were applied to detect and classify these patterns. To simplify the pattern detection, each pattern was searched only within its predefined region of interest, which corresponded to the expected location in the template ROCF. This was possible thanks to the image homography performed during preprocessing. In corresponding regions, the probabilistic Hough transform algorithm was used for line detection, while the contour detection technique was used for shape recognition. If a pattern was present in the drawing, topological analysis was then used to distinguish between the correct, distorted, and misplaced categories.

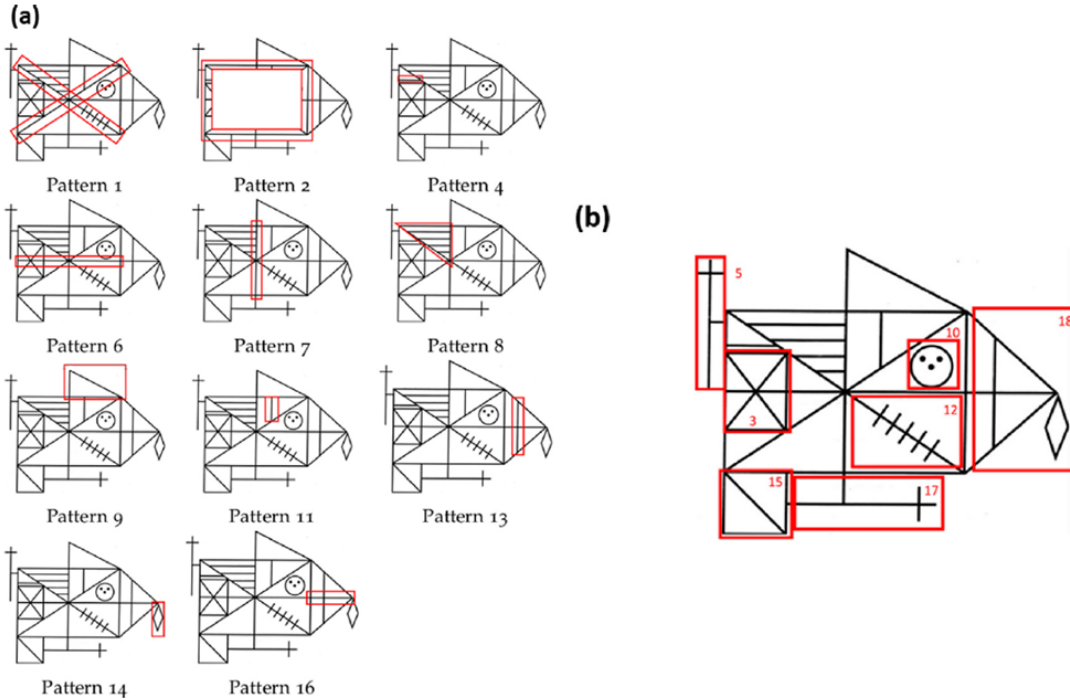


Figure 1.4: Division of 18 ROCF elements into simple and complex patterns, (A) shows simple patterns, (B) contains complex patterns, Source: [6].

On the contrary, complex patterns could not be reliably detected using conventional computer vision methods and therefore required the use of deep learning techniques. A modified ResNet50V2 models were trained to generate simplified feature represen-

tations, also called embeddings, of the patterns in the drawings. During training, the networks were encouraged to map similar elements close to each other in the embedding space, while those that were not similar were pushed further apart. These feature vectors were then used to train two linear support vector machine (SVM) classifiers. The first one to determine whether a pattern was omitted or present, and the second one to decide whether the detected pattern was correct or distorted. For patterns that were not omitted, topological analysis was additionally applied to determine whether they were misplaced [6].

The last phase concentrated on the diagnosis prediction of the subject based on their rocf reproduction, where the classification into one of three categories (healthy, MCI, or dementia) was based on a machine learning model trained on the 18 qualitative predictions of the ROCF elements obtained in the pattern evaluation phase. For this task, a gradient boosting decision tree algorithm (CatBoost) was used to predict the diagnosis. The system achieved an average accuracy of around 70% in classifying the individual patterns. They also achieved strong performance in the overall diagnosis prediction, successfully distinguishing between healthy individuals and those with MCI in 89% of cases, and between healthy and dementia cases with 92% accuracy [6].

1.2.2 Automating Rey Complex Figure Test scoring using a deep learning-based approach: a potential large-scale screening tool for cognitive decline

This recent study, conducted in 2023, differs from the one described in the previous section mainly by using an almost hundred times larger dataset and a method based entirely on deep learning techniques making final prediction based on the whole drawing instead of focusing on individual elements of ROCF. Their training dataset consisted of 20,040 scanned ROCF drawings. These drawings were produced by 6,680 participants, each completing the copy, immediate, and delayed recall tasks. Most participants were without cognitively deficit, about one third had mild cognitive impairment, and a few hundred were diagnosed with dementia. The images were annotated by 32 clinical psychologists. Unlike the previous study, which aimed to directly predict the diagnosis from the reproduced figure, their goal was to design a system that is capable of automatically predicting the total score (ranging from 0 to 36) in the same way as expert would.

Before being fed into the deep learning model, each image was first pre-processed. Noise was reduced using median filtering, and to binarize the images the adaptive thresholding method was used. Subsequently, in order to correct rotation and detect how much the drawing is skewed they utilized the projection profile method. Finally, unnecessary background elements were cropped, the images were resized to 512×512

pixels, and converted to three-channel RGB format.

They used DenseNet architecture pretrained on ImageNet database as the backbone model, which is a specific type of convolutional neural network (CNN). The DenseNet architecture consisted of an one convolutional block, followed by four dense blocks, transition layers with batch normalization, a global average pooling layer, and three fully connected layers ending with a final linear activation function that produced the predicted score. The model was trained using the smooth L1 loss function [13].

They trained their DenseNet model twice using fivefold cross-validation, meaning that the model was trained five times on different subsets of the data, each time reserving one fifth of the training set for validation. The Adam optimizer was used for parameter optimization. The first model was trained on the original dataset, and its training performance was evaluated using the mean absolute error (MAE) and the coefficient of determination (R-squared). The model achieved an MAE of 1.24, which means that its predictions differed from the correct ground truth scores by about one point on average, and an R-squared value of 0.977.

Before training the second and final model, the authors identified images where the predicted scores deviated from the annotated ones by five points or more and analyzed the causes of these errors. Among them, they discovered 80 images, which had quality issues that could be corrected. Some errors were caused by poor image quality caused by scanning, others resulted from mistakes made by psychologists during scoring or were caused by accidental errors made when digitalizing the annotations. These problems were corrected in order to improve the quality of the dataset. Afterwards, the refined dataset was then used to train the final model using the same procedure as before. This second model outperformed the first one, achieving an improved MAE of 0.95 and an increased R-squared of 0.986 [13].

They also used an independent test set containing 150 images used for external validation of the final model. In this evaluation, they analysed how much the predicted scores of the model were in agreement with the scores assigned by five experienced human experts. The model achieved an MAE of 0.64 on the test set, and the average correlation coefficient (R-squared) between the model and each expert was 0.988, which is slightly higher than the average R-squared of 0.983 observed between two distinct human experts. These results indicate that the model's predictions are able to closely approximate how human experts perform ROCF scoring [13].

1.2.3 Automated scoring of a neuropsychological test

The following work was conducted as part of a master's thesis by Bc. Peter Iványi (2023) at the Slovak University of Technology in Bratislava, Faculty of Electrical Engineering and Information Technology. Our research is related to this thesis, since we

use the same dataset and adopt the same score classification system proposed in their work. The aim of Iványi’s thesis was to develop a user interface tool that could automatically classify ROCF drawings into one of four categories using a convolutional neural network. Since the dataset was relatively small and imbalanced, the author proposed dividing the drawings into four classes based on score intervals to achieve more comparable sizes of classes. The defined score intervals for each class are shown in Table 1.2. Since high-scoring images appear more frequently while very low-scoring ones are severely underrepresented, the first class spans an interval that is twice as large as the intervals of the remaining classes.

Their proposed preprocessing procedure involved resizing the images to 500×500 pixels, converting them to Greyscale, and then binarizing them using the Floyd–Steinberg dithering algorithm, which maps each pixel to the most similar color within a defined palette. They designed a simple custom CNN architecture consisting of 10 convolutional layers, combined with max pooling and batch normalization layers, followed by 4 fully connected layers. The ReLU activation function was applied to introduce non-linearity, and a final softmax layer was used for classification into four classes. The training process was repeated multiple times with identical settings to obtain the model with the highest validation accuracy.

The final model achieved 72.5% accuracy on the testing dataset. Further analysis showed that the model most accurately classified images from classes 1 and 4, the lowest and highest scoring groups. Another notable finding was that all misclassified images were predicted as one of the neighboring classes, meaning each incorrect prediction differed from the true class by only a single category. [10].

Class	Score interval	Number of images in class
1	0 - 14.5	240
2	15 - 22	248
3	22.5 - 30	251
4	30.5 - 36	266

Table 1.2: Division of images into four classes based on score intervals [10].

1.2.4 Classification based on Rey figures

The master’s thesis authored by Bc. Miroslav Čobrda in 2024 represented a natural continuation of the previously mentioned work. It utilized the same training software and CNN architecture, which was further improved by adding dropout layers, as well as the same dataset and identical definition of four classes based on score intervals.

The primary goal of the thesis was to build upon the earlier research by exploring the impact of different preprocessing methods and inclusion of data augmentation.

Data augmentation was used to enlarge the dataset and introduce greater variability, by creating copy of each training image and rotating it by 5 degrees. Moreover, two preprocessing strategies were evaluated: use of greyscale images or binarized images using adaptive thresholding. These were compared with Iványi’s preprocessing pipeline, which binarized images using Floyd–Steinberg dithering [10]. That method resulted in suboptimal binarization, producing very thin and pixelized lines, whereas adaptive thresholding generated more natural, smoother strokes.

Performance results for each model variant are shown in Table 1.3. Interestingly, the results indicate that using original greyscale images without any additional preprocessing yields higher accuracy compared to binarized images, which contradicts the assumptions of all other previous research discussed in this chapter, as those studies relied on binarized inputs. The best performance was achieved when greyscale images were combined with rotational augmentation, reaching an accuracy of 92.5%. To verify the reliability of this configuration, the same preprocessing and augmentation setup was evaluated using fivefold cross-validation and the resulting average accuracy was 88.59% [17]. These findings motivated us to attempt to discover augmentation techniques best suited for this dataset and to enhance greyscale images with additional preprocessing aimed at reducing noise and improving the visibility of drawing strokes.

Method	Average accuracy [%]
Floyd-Steinberg dithering without augmentation	72.5%
Adaptive thresholding without augmentation	79%
Greyscale without augmentation	84%
Greyscale with augmentation	92.5%
5-Fold CV, greyscale without augmentation	68.25%
5-Fold CV, greyscale with augmentation	88.59%
10-Fold CV, greyscale without augmentation	73.02%
10-Fold CV, greyscale with augmentation	87.08%

Table 1.3: Average accuracy of CNN with various preprocessing and augmentation methods, source: [17].

Chapter 2

Dataset

The images in the dataset used as the core part of this thesis were collected and evaluated as part of the Neuropsychology project conducted at the Slovak institute Centrum MEMORY [8]. The Neuropsychology project aimed to standardize the evaluation process of 18 neuropsychological tests for clinical use in Slovakia [11]. The images made available to us were obtained from administering the ROCF test to around 350 individuals, evaluated by nine experienced psychologists between 2016 and 2020. Most participants were instructed to draw the complex figure three times: once during the copy phase and twice from memory, after three minutes as the immediate recall and after thirty minutes as the delayed recall.

The original drawings were made with pencil or pen on white paper and were later digitized by scanning. These scans were manually cropped to include only the ROCF drawing, resulting in images of size 1500×1500 pixels, in order to remove unnecessary background and the header of the paper [10]. However, not all drawings produced within the Neuropsychology project were suitable for training machine learning models due to quality issues and therefore had to be excluded from the dataset. As a result, the final dataset contains 1041 images. For each image, three types of information are available: a unique image name, the phase in which it was produced (copy, immediate recall, or delayed recall), and the final total score from evaluation.

The evaluation was done using Osterrieth's scoring system, which allows each drawing to be assigned a score in the range from 0 to 36, with a sensitivity of 0.5 points. Out of all images, 933 belong to the so-called control group, meaning they were drawn by cognitively healthy participants, while the remaining 108 drawings were created by individuals with various types of cognitive dysfunctions who were part of the clinical group.

When it comes to the statistics about the phases of the ROCF test during which the images were created, our dataset is quite balanced, with a slightly higher number of samples from the first and second phases. The exact numbers are as follows: 362 images

from the copy phase, 349 from the immediate recall phase, and 330 from the delayed recall phase (see histogram in Figure 2.1). Regarding the distribution of images across the 73 possible scores (ranging from 0 to 36 with a sensitivity of 0.5), we observed that higher scores tend to be more frequent (see histogram of score frequencies in Figure 2.2). Notably, drawings with scores up to 6 represent only 4.51% of the dataset, whereas those with scores between 30 and 36 make up 25.07%. The percentage distribution of images in six-point intervals is shown in Table 2.1.

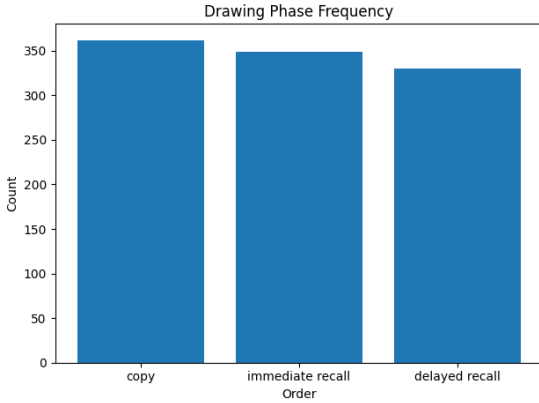


Figure 2.1: Distribution of images across the three ROCF phases.

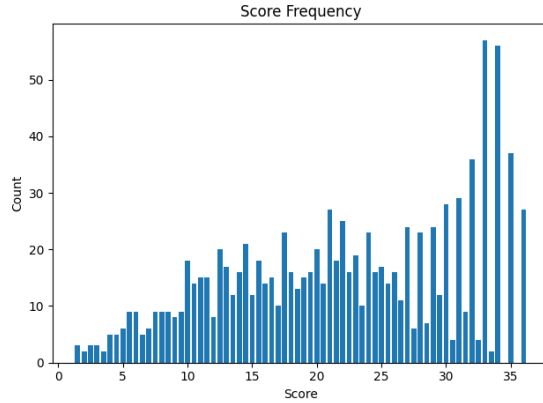


Figure 2.2: Distribution of images by total ROCF score.

Furthermore, some scores do not appear in the dataset at all: specifically 0, 0.5, 1, 34.5, and 35.5. There also seems to be a slight preference for whole-number scores, since drawings with scores with half a point (.5) represent only about 35% of the dataset. In conclusion, the distribution of scores is not evenly spread out, and there appears to be a certain bias toward higher whole-number scores.

Score interval	Percentage of images
(0, 6)	4.51%
(6, 12)	12.01%
(12, 18)	18.64%
(18, 24)	20.75%
(24, 30)	19.02%
(30, 36)	25.07%

Table 2.1: Percentage distribution of image scores in six-point intervals.

An example of a ROCF drawing with a high score from our dataset is shown in Figure 2.3. However, due to the nature of the ROCF task, the drawings in our dataset show a great deal of variety and can be quite challenging for computer vision processing. Common issues that appear in the dataset include lightly drawn lines that are not continuous or appear choppy (see Figure 2.4), an uneven or shaky lines caused by trembling hands, and drawings that are deformed or distorted (see Figure 2.5). In

some cases, the images contain many missing elements or so many changes that they barely resemble the original ROCF figure (see Figure 2.6).

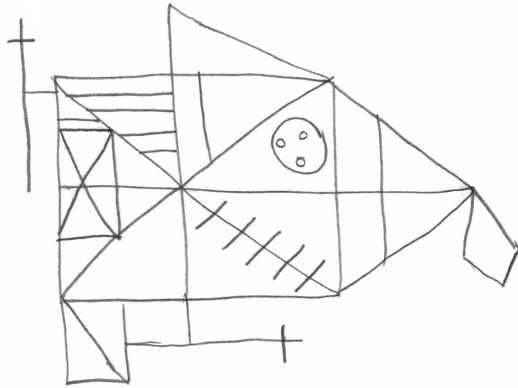


Figure 2.3: Example of a high-quality drawing with a score of 35. Source: [8]

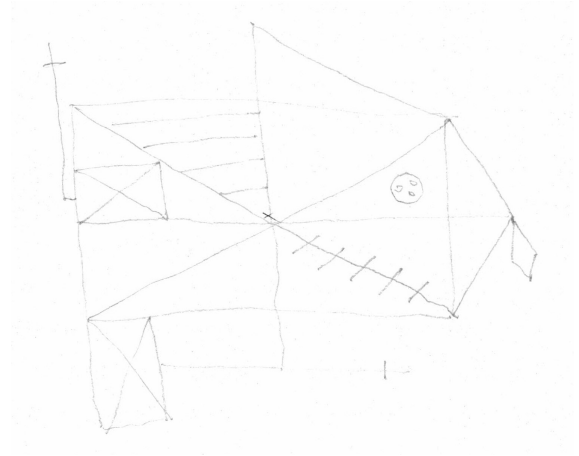


Figure 2.4: Drawing characterized by light and discontinuous lines, with a score of 24.5. Source: [8]

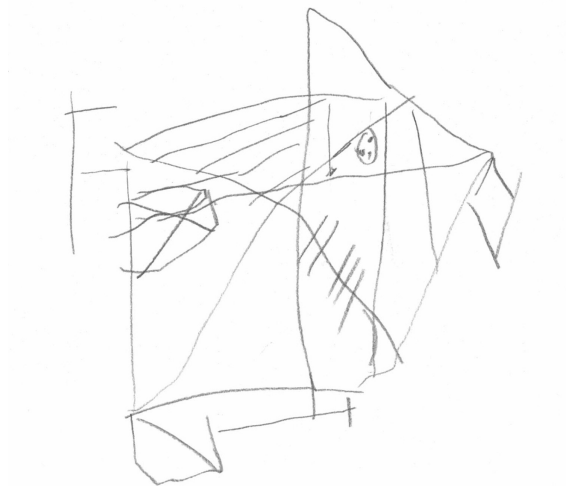


Figure 2.5: Drawing showing strong distortion of the original figure, with a score of 20.5. Source: [8]

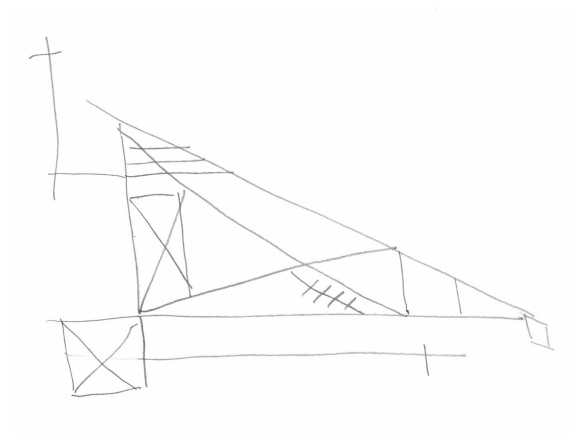


Figure 2.6: Drawing that only slightly resembles the original ROCF, with a score of 7. Source: [8]

Chapter 3

Objective and Methodology

goal of the thesis - mention every goal in the annotation from ais - subchapter for each goal what is my approach - mention article on which my methodology is based upon - my proposal of the solution

Chapter 4

Software design

short report - what inputs, outputs, UML diagrams, all main blocks/components of the pipeline - basic principles

Chapter 5

Implementation

text

Chapter 6

Research

TODO: validation, and results, graphs, images (described and explained), why we are better, false positives, false negatives, why are we not successful, changes of parameters of the system and their evaluation in a table

Draft of the research chapter

In this thesis, the dataset is divided into fixed training, validation, and test sets, each maintaining the same proportional representation of all classes. These splits are consistently used for every model to ensure fair and reproducible comparisons during training and validation.

The performance of all models is evaluated using standard metrics: recall, precision, F1 score, mean absolute error (MAE), and confusion matrix. The results were obtained using test-time augmentation to improve the accuracy. Test time augmentation: for each image, five versions are sent to the model—one unmodified, one rotated, one translated, one with altered brightness/contrast, and one combining all three augmentations. The final prediction is the majority vote across the five outputs.

Initially, two deep learning architectures are compared: the ResNet-18 convolutional neural network and the Swin Transformer. Both models are trained on the same training dataset and evaluated on the same validation and test datasets without any preprocessing or data augmentation. The experiments are conducted with simplified settings: input image size of 400×400 pixels and a four-class classification setup proposed By Ivanyi [10]. Their performance is compared with a baseline method proposed by Iványi et al., which uses a simple convolutional neural network.

Subsequently, both ResNet-18 and Swin Transformer models are trained again using five different data augmentation techniques. Their results are compared with the baseline method from Iványi and with the initial non-augmented versions of both models. Based on these comparisons, the best-performing architecture and augmentation strategy are selected. Preliminary results indicate that ResNet-18 achieves better performance, and the most effective augmentation technique is a combination of random

TTA metrics	Ivanyi	ResNet18	Swin Transformer
recall	72.5%	71.43%	68.57%
precision		72.14%	68.61%
F1 score		70.51%	68.25%
loss		0.9951	1.0253

Table 6.1: Test-time augmentation results on test set of ResNet18 and Swin Transformer trained without augmentation on 4-class classification and results reported by Ivanyi: [10]

translation, rotation, and adjustments of color brightness and contrast.

The best-performing configuration (ResNet-18 with optimal augmentation) will then be trained from scratch for a six-class classification task, where each class corresponds to a six-point interval of possible ROCF scores. The input image size will be increased to 500×500 pixels. Four variations of the ResNet-18 model will be trained:

- Without preprocessing and augmentation (serving as a new baseline ground truth method),
- With the best augmentation and no preprocessing,
- With no augmentation and with preprocessing,
- With both preprocessing and the best augmentation.

In addition, an encoder–decoder architecture will be trained on the same dataset. The encoder will be used to extract low-level feature vectors from the drawings, which will serve as input to a simple feedforward neural network performing the final classification. This encoder–neural network approach will be compared with the four ResNet-18 models trained for the six-class task.

TTA Metric	Augmentation type					
	No	contrast, brightness	translation	rotation	crop	contrast, brightness, translation, rotation
recall	71.43%	72.38%	73.33%	73.33%	50.48%	77.14%
precision	72.14%	72.69%	73.51%	73.37%	53.43%	77.61%
F1 score	70.51%	71.58%	73.17%	72.77%	42.88%	76.94%
loss	0.9951	0.8071	0.9922	0.9214	1.0312	0.6135

Table 6.2: Test-time augmentation results on test set of ResNet18 comparing multiple augmentation techniques on 4-class classification

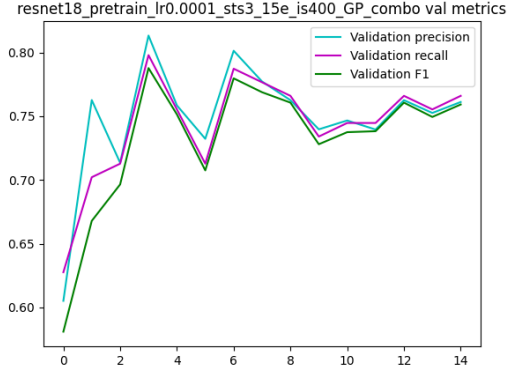


Figure 6.1: Performance metrics on validation set computed during training of ResNet-18 using combined augmentation

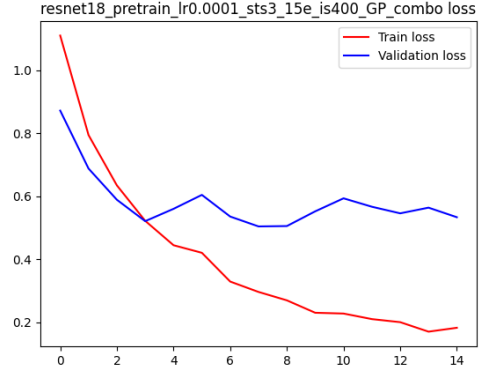


Figure 6.2: Loss on validation and training set computed during training of ResNet-18 using combined augmentation

	Augmentation type					
TTA Metric	No	contrast, brightness	translation	rotation	crop	contrast, brightness, translation, rotation
recall	68.57%	64.76%	66.67%	73.33%	24.76%	69.52%
precision	68.61%	65.41%	66.71%	73.29%	6.13%	69.08%
F1 score	68.25%	64.98%	66.57%	72.23%	9.83%	68.85%
loss	1.0253	0.987	0.7173	0.7681	2.4093	0.7053

Table 6.3: Test-time augmentation results on test set of Swin Transformer comparing multiple augmentation techniques on 4-class classification

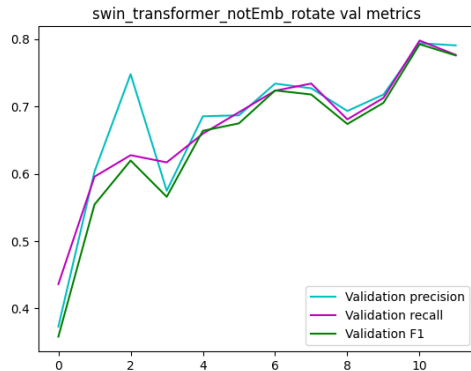


Figure 6.3: Performance metrics on validation set computed during training of Swin Transformer using rotational augmentation

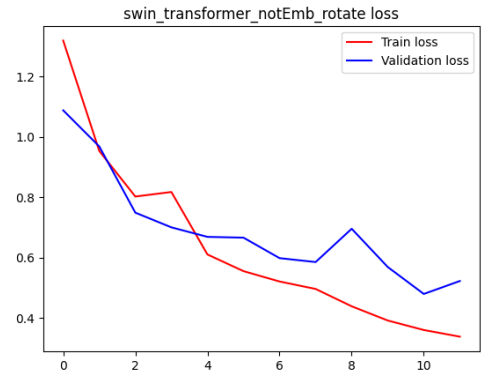


Figure 6.4: Loss on validation and training set computed during training of Swin Transformer using rotational augmentation

Chapter 7

Results

interpretation of the results, with input from experts in the field, if the results are good or bad, limits of the method, what are the constraints, up to what point does it work (under what circumstances/cases does it work)

7.1 Summary

Summary text

Chapter 8

User Manual

short description of how to use the software;

Chapter 9

Conclusion

conclusion text

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